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Local opportunity does not ensure end-to-end gain

Observed case	Favorable local measurement		Realized outcome
M4 · HumanEval30 Preregistered, fixed k = 2	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + profitable** 1.387× · 90/90 exact
M4 · representative 24Q Frozen representative study	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + negative** 0.842× · 72/72 exact
RTX 5070 Ti Laptop CUDA batched verification	$V_2^{\text{micro}} = 0.983–1.009$ Near one target-decode step	→	□ **Correctness blocked** No admissible speedup
358H · NPU draft Qualified Phase-3 path	48.08 vs 30.81 tokens/s NPU vs CPU · isolated, 8,192 tokens	→	● **Exact + negative** 0.352× · 90/90 exact

Categorical cases, not a hardware ranking. Filled = qualified; open = blocked. Details and CIs: Figures 2–4.

Micro cost ≠ complete target work; isolated throughput ≠ stateful cost. Speedups cover post-prefill decode.

Figure 1: Favorable local signals, divergent realized outcomes. Near-one-step target batching and favorable isolated drafting do not consistently translate into qualified end-to-end gains in the tested cases. The same tested M4 Metal stack exhibits a profitable HumanEval regime and a negative representative-24Q regime in separate frozen studies. RTX batched verification is physically cheap but fails correctness qualification. The 358H NPU has higher isolated median throughput at 8,192 tokens yet reaches only 0.352× realized speedup. Rows are heterogeneous case summaries, not a hardware ranking or common quantitative axis. M4 micro values describe the shared path in a separate micro study. Headline speedups cover post-prefill decode; Figures 2–4 give conditions, uncertainty and the two measured diagnoses.

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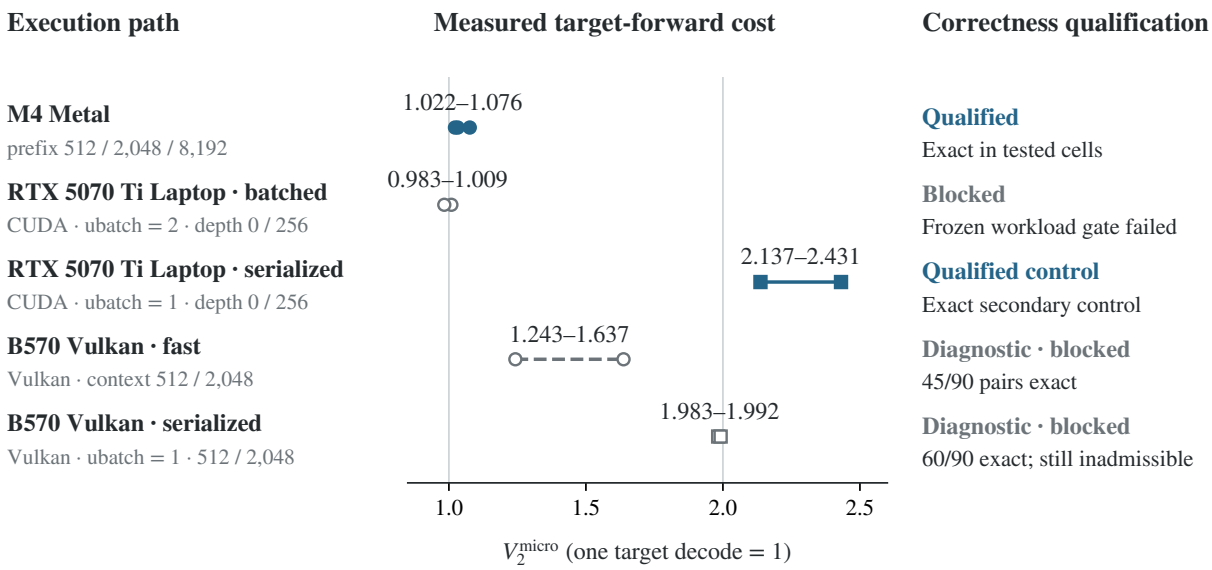


Figure 2: Correctness qualification changes which verification path is usable. Each mark is one measured context cell; range segments span those cells, not confidence intervals. Filled markers indicate qualified tested paths; open markers indicate blocked or diagnostic paths. Serialization yields an exact secondary RTX control, but the B570 serialized diagnostic remains globally inadmissible despite improving observed exactness from 45/90 to 60/90 pairs. Contexts and runtime stacks are not fully matched. The target-forward proxy V_2^{micro} excludes complete realized target-side work R_{target} and is not an end-to-end speedup. Qualification is scoped to the frozen workload and execution path, not a device family.

Figure-only narrative proof; native MLSys width. Print at 100%.

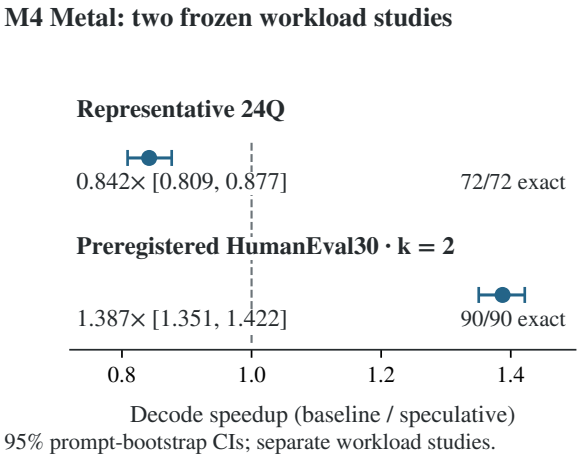
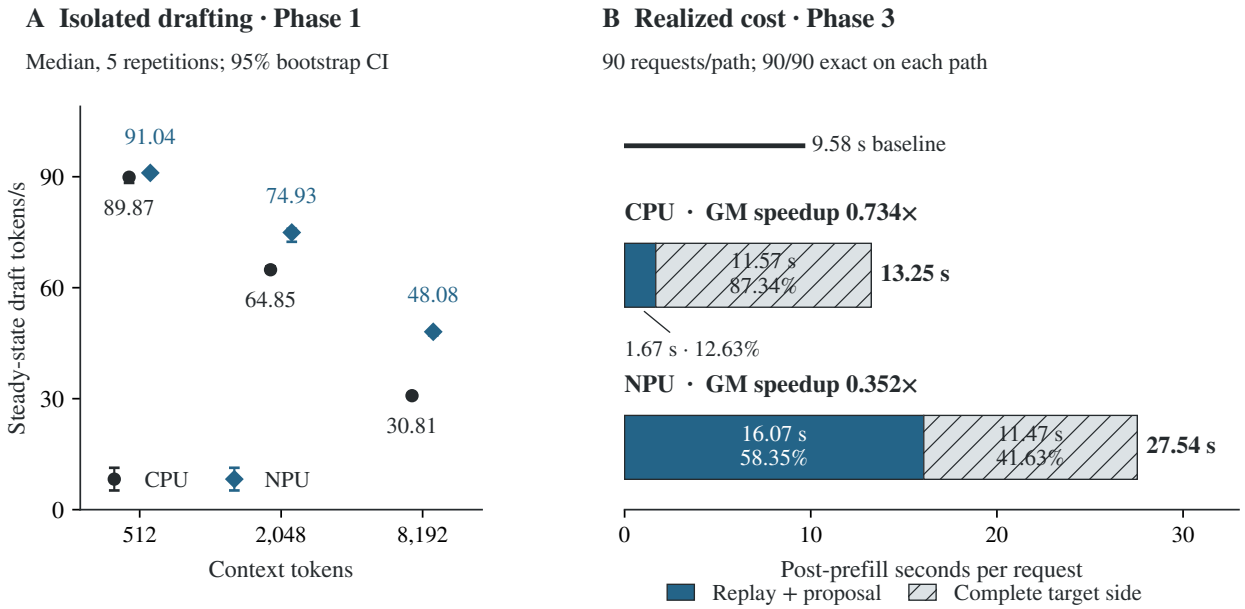


Figure 3: Opposite profitability regimes on the same tested M4 stack. Representative 24Q yields 0.842× [0.809, 0.877], while the separately preregistered HumanEval30 fixed-$k = 2$ study yields 1.387× [1.351, 1.422]. Points are campaign prompt geometric means of paired decode speedups; intervals are campaign-specific 95% prompt-bootstrap intervals. The dashed line marks parity. Both paths satisfy token equality (72/72 and 90/90 pairs), which is not functional HumanEval pass@k. These are separate studies, not a paired between-workload comparison; differing workloads and session conditions do not isolate workload-only causality.

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Residual: CPU 4.81 ms (0.04%); NPU 5.06 ms (0.02%). Bars: arithmetic means; speedups: prompt GMs.

Replay and proposal share one timed call. Phase-1 steady-state throughput excludes the full-prefix state cost.

Speedup 95% CIs: CPU [0.693, 0.771]; NPU [0.333, 0.369]. Initial prompt prefill is excluded from Phase 3.

Figure 4: Isolated draft advantage does not survive the measured stateful integration. (A) Phase-1 median steady-state draft throughput and 95% repetition-bootstrap intervals, five repetitions per cell; the NPU median advantage grows across the tested contexts. (B) Phase-3 arithmetic mean post-prefill seconds per request over 90 observations per path, all exact. CPU and NPU totals are 13.25 s and 27.54 s against a 9.58 s mean baseline. The CPU path is target-side dominated (87.34%); the NPU path spends 58.35% in the measured replay-plus-proposal call. The 0.734× and 0.352× headlines use the separate prompt-GM estimator and are not ratios of the arithmetic mean bars. Replay and proposal cannot be separated within this timer. Initial prompt prefill is excluded. This diagnoses the tested runtime interface, not intrinsic NPU capability or a measured benefit from a state-persistent intervention. Together with Figure 2, it identifies two measured reasons local opportunity need not become realized gain; the M4 contrast remains a separate empirical observation.

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Categorical regimes under the frozen correctness contract

Mechanism	Not admissible	Correctness-qualified	
	No admissible speed claim	Negative ($S < 1$)	Profitable ($S > 1$)
Workload conditioning		● **M4 representative 24Q [P]** 72/72 exact	● **M4 HumanEval30 [P]** 90/90 exact
Verification constraint	○ **RTX batched [M]** Cheap micro path; gate fails	● **RTX serialized [S]** CPU→RTX / RTX→RTX; exact	
Correctness barrier	○ B570 fast: 45/90 [D] ○ B570 serialized: 60/90 [D]		
Target-side cost		● **358H CPU [P/M]** 90/90; target side 87.34%	
State-interface cost		● **358H NPU [P/M]** 90/90; replay + proposal 58.35%	

P primary · S secondary · M mechanism · D diagnostic. Fractions count exact token-ID pairs.

Figure A1: Evidence inventory across tested regimes. Categorical columns distinguish globally blocked paths from exact negative and profitable regimes. Rows annotate observed constraints rather than define an exhaustive taxonomy. Both B570 paths remain diagnostic and globally inadmissible. RTX serialized controls are secondary evidence; 358H outcome and decomposition support primary and mechanism claims, respectively. Empty cells indicate no displayed case, not impossibility. No hardware ordering or continuous correctness metric is implied.